# SAYAN BANERJEE

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## EDUCATION

**Banaras Hindu University** 2024 – 2026
*Masters of Science in Statistics and Computing* CGPA: 8.5

**Sister Nivedita University** 2021 – 2024
*Bachelor of Science in Statistics (Hons)* CGPA: 8.51

## EXPERIENCE

### Subject Matter Expert Link

**Chegg Inc**

Oct 2022 – Dec 2023 Remote

- Served as a Subject Matter Expert in Probability and Statistics, providing high-quality content and solutions to students , solved 689 questions delivering quality solutions.

### AI/ML Intern Link

**CodimAI**

July 2025 – Current Mumbai,India

- Developed a speech-to-speech Diet Planner, and a Lab Report Reader, alongside conducting data analysis to support product innovation.

## TECHNICAL SKILLS

**Languages and Tools:**

- SQL, Python , R , PowerBI, Excel (Basic), Git, Streamlit , MS Word & Powerpoint

**Libraries and Frameworks:**

- Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, Flask, TensorFlow, Scipy , Matplotlib , Seaborn

**Statistical and ML Concepts:**

- Hypothesis testing , Survival Analysis , Parametric & Non-Parametric Inference , Sampling Theory , Regression & Classification Models , Clustering and Segmentation Models

## CERTIFICATIONS

- Data Analytics with Python | *NPTEL* (All India Rank 43 out of 10022 Candidates) Link
- Leetcode SQL 50 | *Leetcode* Link
- Deep Learning - IIT Ropar | *NPTEL* (All India Rank 11 out of 4869 candidates) Link
- Summer Analytics 2025 | *IIT Guwahati* Link
- Data Science Simulation | *BCG Forage* Link

## ACADEMIC ACHIEVMENTS

**IIT JAM MATHEMATICAL STATISTICS '24 Link**

- Qualified with a rank of 88 , percentile: 97.51

**GATE STATISTICS '24 Link**

- Qualified with a rank of 125 , percentile: 92.62

## PROJECTS

**Swiggy Delivery Time Prediction Link | Demo**

*Linear Regression ,Random Forest, Decision Tree, SVM, Ensemble models , XgBoost , LightBGM, Optuna , ANN*

- Built a high-performance regression system on 40K Kaggle dataset for delivery time prediction; conducted data preprocessing, EDA, feature engineering, feature selection (Forward Selection, RFE, VIF , Lasso ,Ensemble feature importance).Trained and optimized multiple models(ML and DL) achieving $R^2$ = 0.88 with a Voting Regressor (XGBoost + LightGBM)

**Multiple Disease Prediction System Link | Demo**

*Logistic Regression ,Random Forest, SVM, KNN , Ensemble models , XgBoost , CNN Architectures , SHAP*

- Developed an integrated Multiple Disease Prediction System using Machine Learning and Deep Learning for early diagnosis of Heart Disease, Diabetes, Pneumonia, and Diabetic Retinopathy. Combined tabular clinical data and medical imaging to deliver accurate, interpretable predictions via an interactive Streamlit app. Implemented Random Forest, SVM, Decision Tree, CNN, and UNet models using Scikit-learn and TensorFlow/Keras, with SHAP integration for model explainability and transparency.

**Churn Survival Modeling: Predicting Customer Lifetimes Link**

*Kaplan-Meier Curves , Cox PH model , Random Forest , CatBoost, Random Survival Forest , AFT Models*

- Developed churn prediction system combining classification models (Logistic Regression, Random Forest, XGBoost, SVM) with survival analysis to identify at-risk customers. Performed Kaplan-Meier analysis with log-rank tests across customer segments. Implemented Cox Proportional Hazards and Accelerated Failure Time (AFT) models to quantify risk factors and estimate lifetimes. Deployed Random Survival Forests and Gradient Boosted Trees for robust predictions (C-index evaluation). Generated personalized survival curves for targeted retention and lifetime value estimation.

**Time Series Forecasting and Automation: US Price Index Analysis Link | App**

*Time Series, Hugging Face Pipeline , Statsmodels , Forecasting , Streamlit*

- Conducted statistical analysis including stationarity tests (ADF, KPSS), seasonal decomposition, and exponential smoothing (Holt-Winters). Quantified long-term memory using Hurst exponent and GPH filter. Evaluated temporal variability through rolling statistics and tested for heteroscedasticity (ARCH tests). Implemented ARFIMA, SARIMA, LSTM, and GRU models with comprehensive diagnostics including residual normality tests (Shapiro-Wilk), autocorrelation analysis (Ljung-Box).